Data processing / preparation for small molecules and peptides

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DIKU, University of Copenhagen · Lecture 13 · AI- & data-driven drug design · 13 November 2026

→ next ← back F fullscreen L laser O overview S speaker notes T dark mode

Outline

1. **What is machine learning?**
   a model takes a table of numbers
2. **Representing small molecules and peptides**
   turning a molecule into a format a machine can read
3. **Why let the model learn the representation?**
   learned from data rather than handcrafted: graphs, embeddings, 3D

A classic example: what will this house sell for?

	living area m²	bedrooms	built	sold for US $
1	154	3	1960	$215,000
2	83	2	1961	$105,000
3	196	3	1968	$244,000
4	124	2	2001	$213,500
5	305	4	2003	$538,000
new	151	3	1997	?

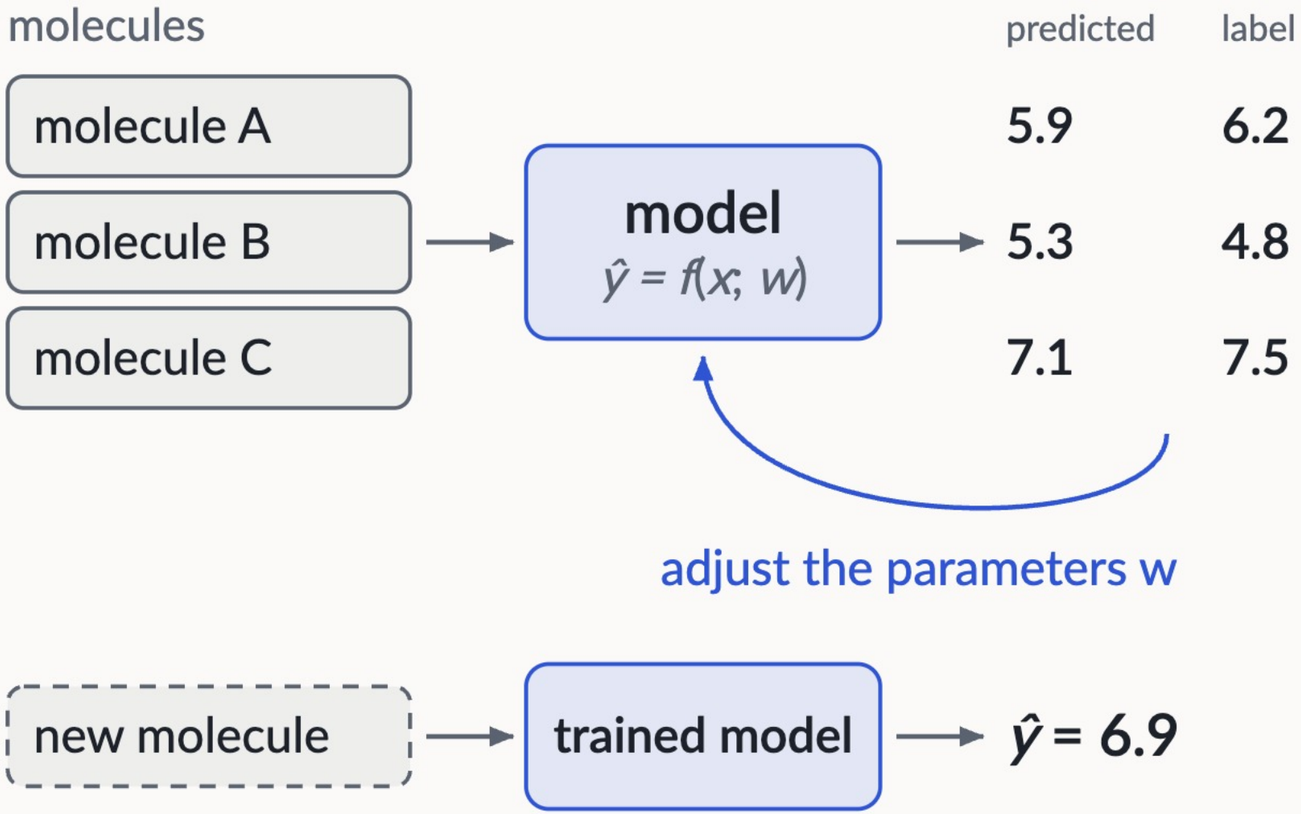
1. Houses that were sold
2. A new house: what price?
3. Learn the rule from examples
4. Our houses are molecules

a house → **a molecule**

area, bedrooms, year built → **???**

sale price → **activity, solubility, ...**

What is machine learning?



1 **A dataset**

molecules with labels

2 **A model**

parameters that make predictions

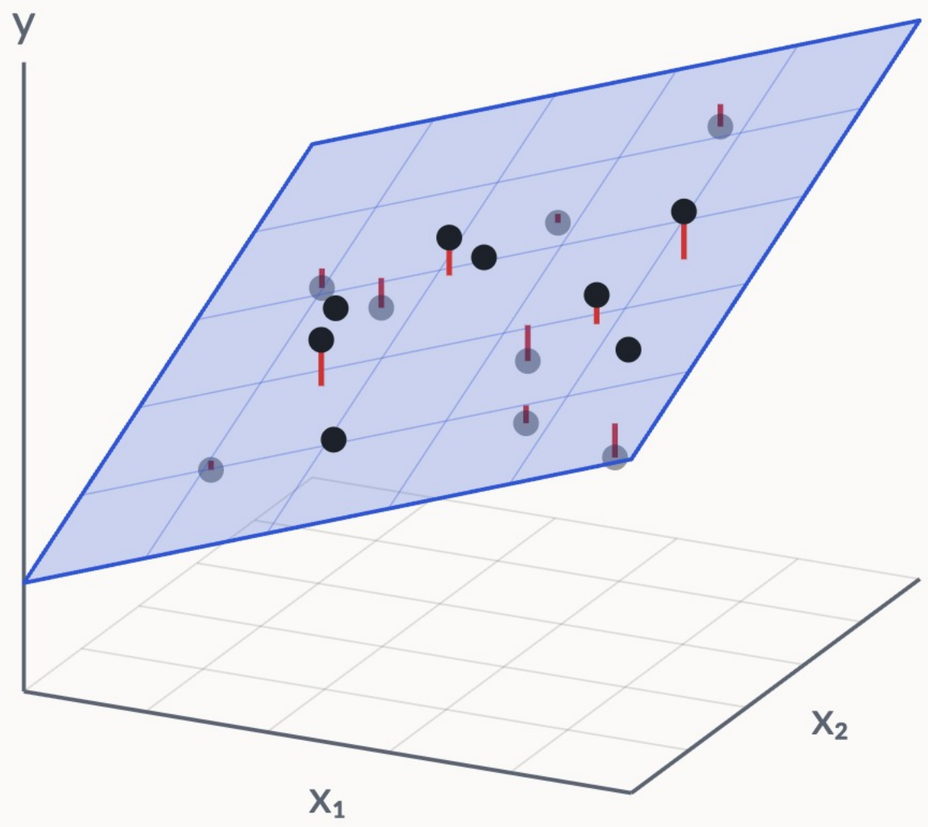
3 **Training**

tune the parameters to match the labels

4 **Prediction**

on something new

Linear regression



1. Data: pairs (x, y)
2. Model: a straight line

   $\hat{y} = w \cdot x + b$
3. Training: make the errors small
4. Predict: a new x gives ŷ
5. Two features: a plane

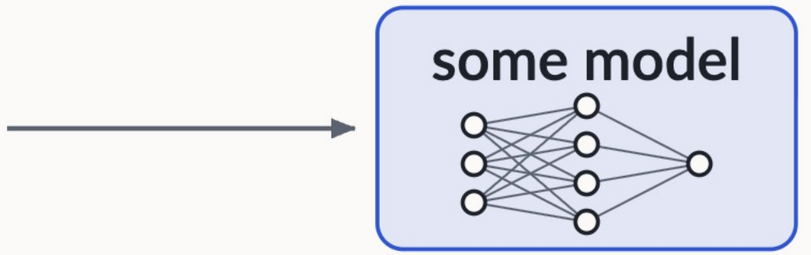
   $\hat{y} = w_1 x_1 + w_2 x_2 + b$
6. More features: still a simple sum

   $\hat{y} = w_1 x_1 + w_2 x_2 + \ldots + w_d x_d + b$

A model takes a table

	x_1	x_2	x_3	x_4
molecule 1	0.8	3	1.2	0
molecule 2	2.4	1	0.3	1
molecule 3	1.1	4	2.7	0
molecule 4	3.0	2	0.9	1

a table

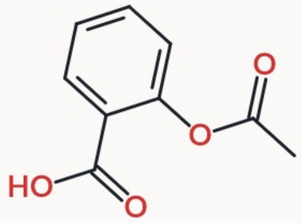


$\hat{y}$
6.2
4.8
7.5
5.1

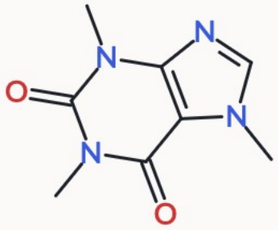
some output

Rows: examples. Columns: features.

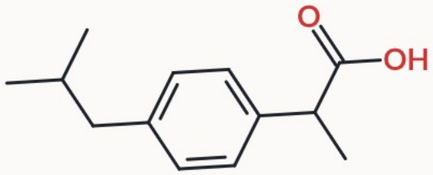
Molecules are not rows



aspirin
13 heavy atoms



caffeine
14 heavy atoms



ibuprofen
15 heavy atoms

data preprocessing
molecule → numbers

some model

$\hat{y}$
6.2
4.8
7.5
5.1

some output

That step is what we cover today.

What you start from: data formats

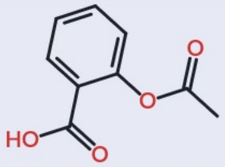
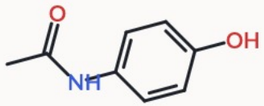
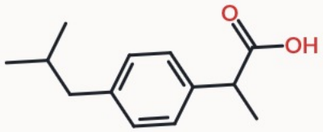
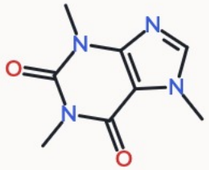
Format	The file	Holds	From
SMILES	`CC(=O)Oc1ccccc1C(=O)O`	atoms and bonds as text	ChEMBL, PubChem

SDF / MOL	`3.4830 −0.6900 −0.2150 C`
	`2.0720 −0.7220 0.2930 C`
	`1.7630 −0.7750 1.4760 O`
	`…`
	`1 2 1 0`
		atoms, coordinates, bonds	PubChem, docking

PDB / mmCIF	`ATOM 1 N MET A 1 27.340 24.430 2.614`	proteins in 3D	PDB, AlphaFold
FASTA	`>GLP-1 (7-36)` `HAEGTFTSDVSSYLEGQAAKEFIAWLVKGR`	residue letters	UniProt

First: parse, strip salts, merge duplicates, check units.

The classic answer: descriptors

		molecular weight	logP	H-bond donors	H-bond acceptors	polar surface	rotatable bonds
	aspirin	180.2	1.31	1	3	63.6	2
	paracetamol	151.2	1.35	2	2	49.3	1
	ibuprofen	206.3	3.07	1	1	37.3	4
	caffeine	194.2	−1.03	0	3	61.8	0

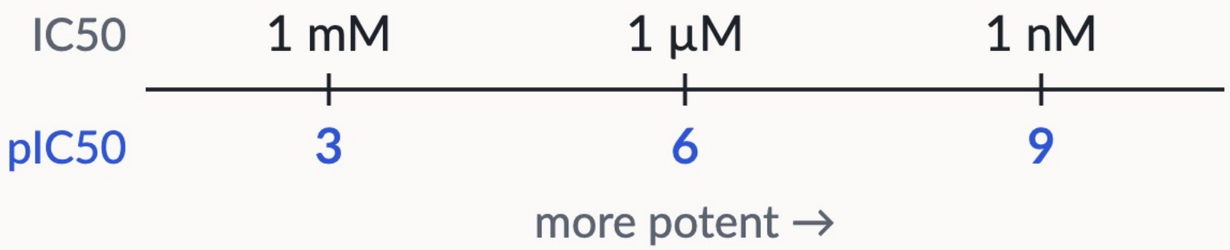
A molecule is now a row. The model sees only what we computed.

Transformations and normalisation

	mol. weight	logP	H-bond donors
aspirin	−0.14	+0.09	+0.00
paracetamol	−1.55	+0.12	+1.41
ibuprofen	+1.14	+1.30	+0.00
caffeine	+0.55	−1.51	−1.41

standardised

the measurement:



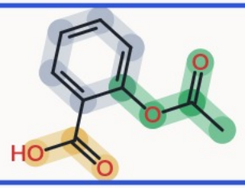
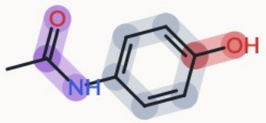
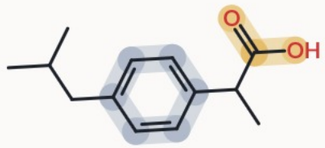
1. **Columns live on different scales**
2. **Standardise each column**

   $z = (x - \text{mean}) / \text{sd}$
3. **Transform skewed measurements**

   $pIC50 = -\log_{10} IC50$
4. **Fit on the training set only**

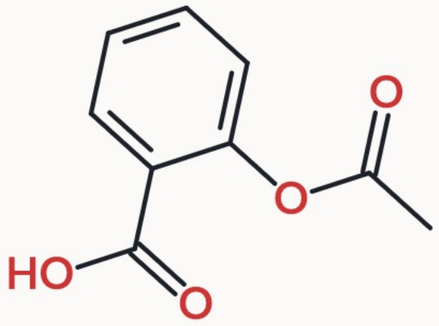
Fingerprints: functional groups

Which groups does it have?

	carboxylic acid	ester	amide C(=O)–N	phenol OH	benzene ring
 aspirin	1	1	0	0	1
 paracetamol	0	0	1	1	1
 ibuprofen	1	0	0	0	1
 caffeine	0	0	1 count: 3	0	0

MACCS keys: 166 such questions. Count instead of flag: a count fingerprint.

Same idea, three names: ECFP, ECFC, FCFP



1 1 0 0 1

functional groups: there or not

ECFP the list writes itself: every small fragment

1 0 1 1 0 0 1 0 1 0 …

ECFC count, not just yes or no

1 0 4 2 0 0 1 0 1 0 …

FCFP atoms by role, not by element

1 1 0 0 1 0 0 1 0 0 …

All three: a row of answers about which substructures are there.

Descriptors or fingerprints?

Descriptors

180.2 | 1.31 | 1 | 3 | 63.6

+ Easy to interpret
- Different molecules, same numbers
- Handcrafted: may miss what matters

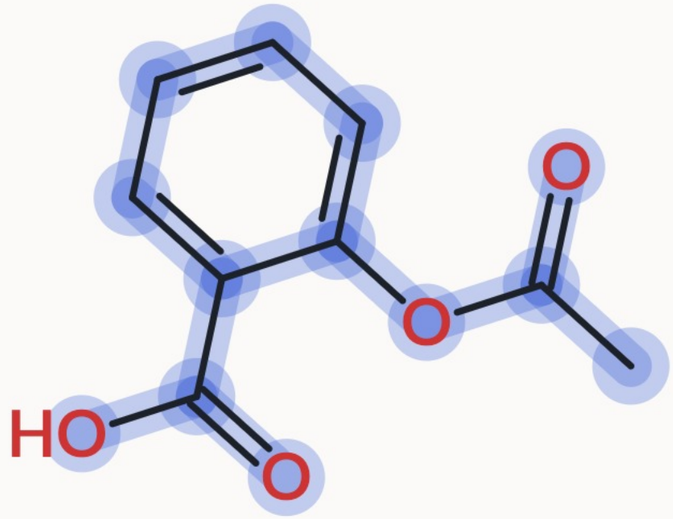
Fingerprints

1 0 1 1 0 0 1 0 1 0 …

+ More structural information
- Molecules can still look identical
- Harder to interpret

Both are designed by hand.

SMILES: writing a molecule as text

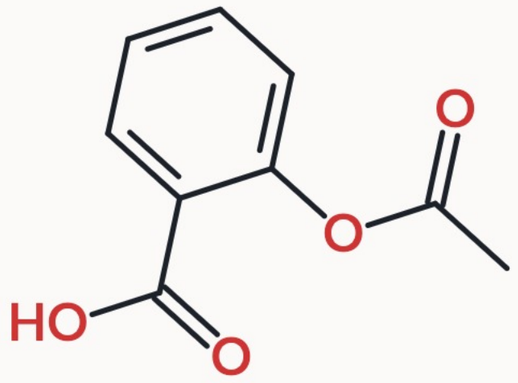


`CC(=O)Oc1ccccc1C(=O)O`

aspirin, in 21 characters

1. **Walk along the atoms**
   hydrogens are left out
2. **Brackets: a side branch**
   = is a double bond
3. **Lower case: aromatic**
   a digit opens a ring
4. **The same digit closes the ring**
5. **Finish: the molecule as text**

Idea #2: SMILES as model input



`CC(=0)0c1ccccc1C(=0)0`

1 Not a fixed size

needs a sequence model

aspirin	`CC(=0)0c1ccccc1C(=0)0`	21 characters
caffeine	`Cn1c(=0)c2c(ncn2C)n(C)c1=0`	26 characters
ibuprofen	`CC(C)Cc1ccc(C(C)C(=0)0)cc1`	26 characters

2 Not unique

one molecule, many spellings

`c1(C(=0)0)c(cccc1)0C(C)=0`	= aspirin
`c1c(C(=0)0)c(0C(C)=0)ccc1`	= aspirin
`c1cccc(c1C(=0)0)0C(=0)C`	= aspirin

Raw characters in: the model must find the chemistry itself.

Step 1: tokenization

C	C	(	=	O	)	O	c	1	c	c	c	c	c	1	C	(	=	O	)	O
1	1	2	3	4	5	4	6	7	6	6	6	6	6	7	1	2	3	4	5	4

Cl	c	1	c	c	c	c	c	1

chlorobenzene: Cl is one token

vocabulary

C	1	(	2	=	3	O	4	)	5	c	6	1	7	n	8	2	9

Vocabulary: every token gets a number.

Step 2: padding

aspirin 21 tokens

C	C	(	=	O	)	O	c	1	c	c	c	c	c	1	C	(	=	O	)	O	·	·	·	·	·
1	1	2	3	4	5	4	6	7	6	6	6	6	6	7	1	2	3	4	5	4	0	0	0	0	0

caffeine 26 tokens

C	n	1	c	(	=	O	)	c	2	c	(	n	c	n	2	C	)	n	(	C	)	c	1	=	O
1	8	7	6	2	3	4	5	6	9	6	2	8	6	8	9	1	5	8	2	1	5	6	7	3	4

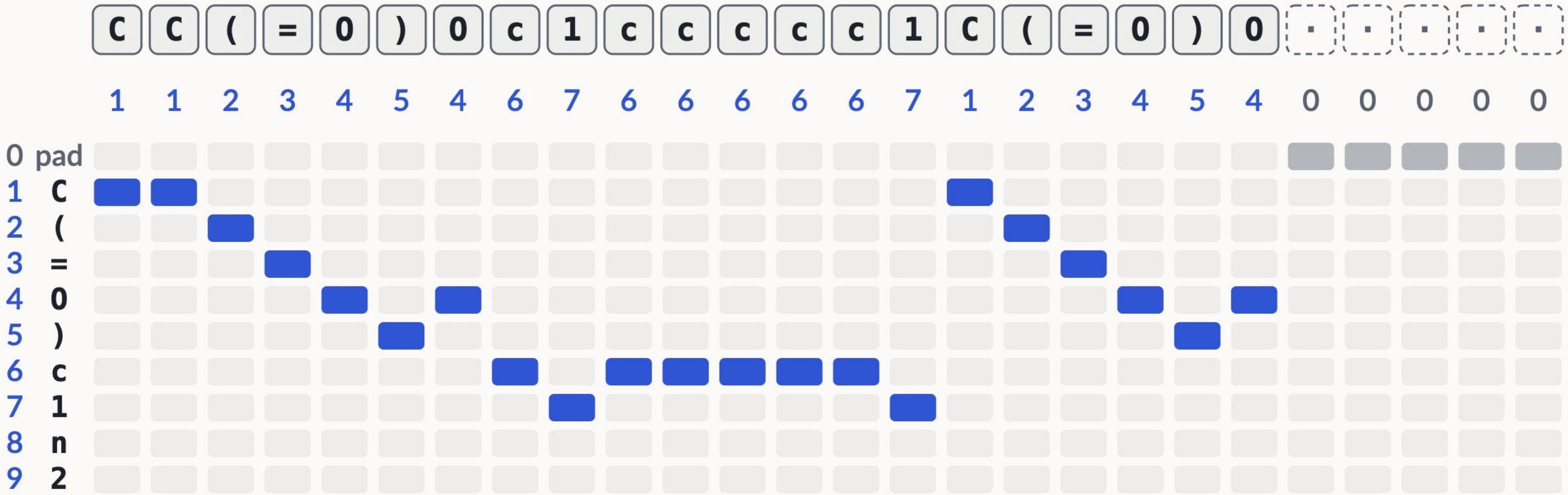
ibuprofen 26 tokens

C	C	(	C	)	C	c	1	c	c	c	(	C	(	C	)	C	(	=	O	)	O	)	c	c	1
1	1	2	1	5	1	6	7	6	6	6	2	1	2	1	5	1	2	3	4	5	4	5	6	6	7

26 positions

Every molecule: the same number of numbers. Padding is 0.

Step 3: one-hot encoding



Every token becomes one column.

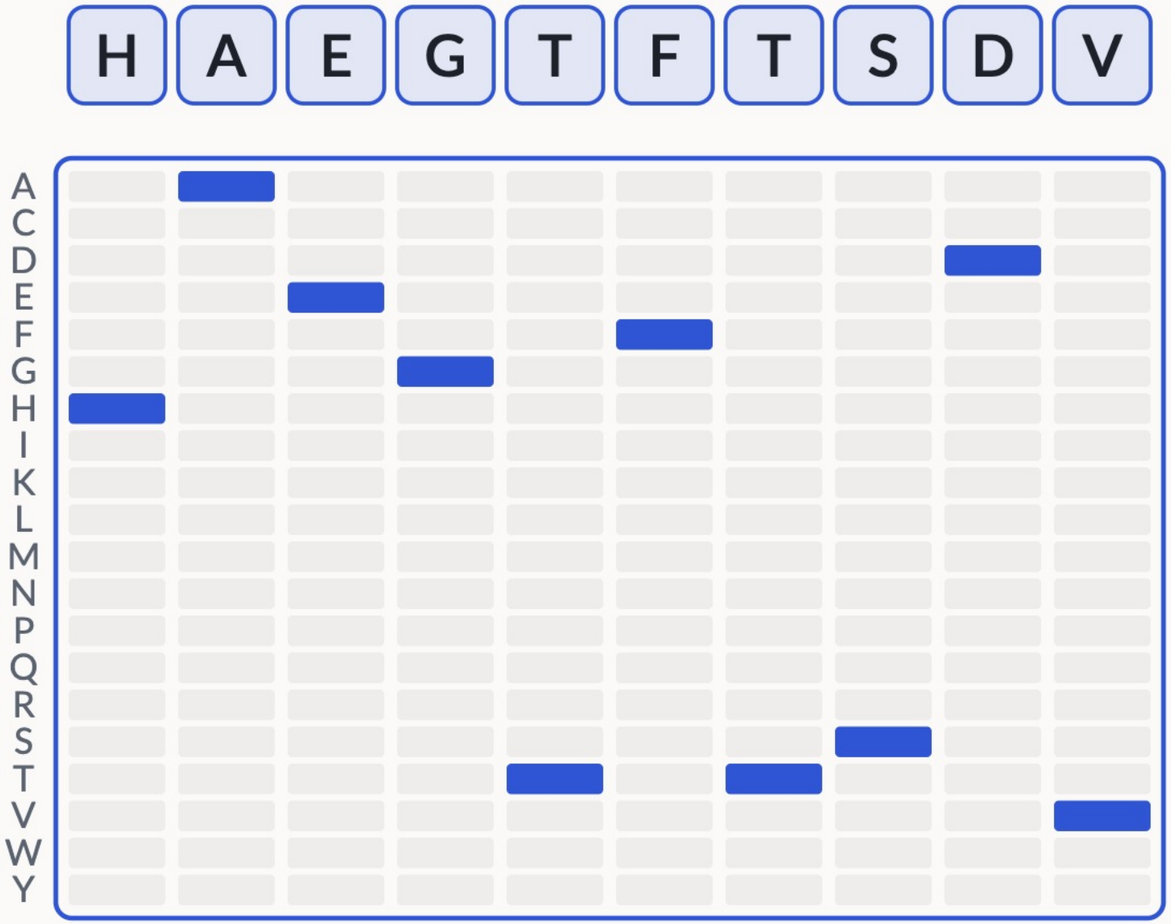
From SMILES to a table

26 columns, every row

aspirin	1	1	2	3	4	5	4	6	7	6	6	6	6	6	7	1	2	3	4	5	4	0	0	0	0	0
caffeine	1	8	7	6	2	3	4	5	6	9	6	2	8	6	8	9	1	5	8	2	1	5	6	7	3	4
ibuprofen	1	1	2	1	5	1	6	7	6	6	6	2	1	2	1	5	1	2	3	4	5	4	5	6	6	7

Every molecule is a row of the same length: a table.

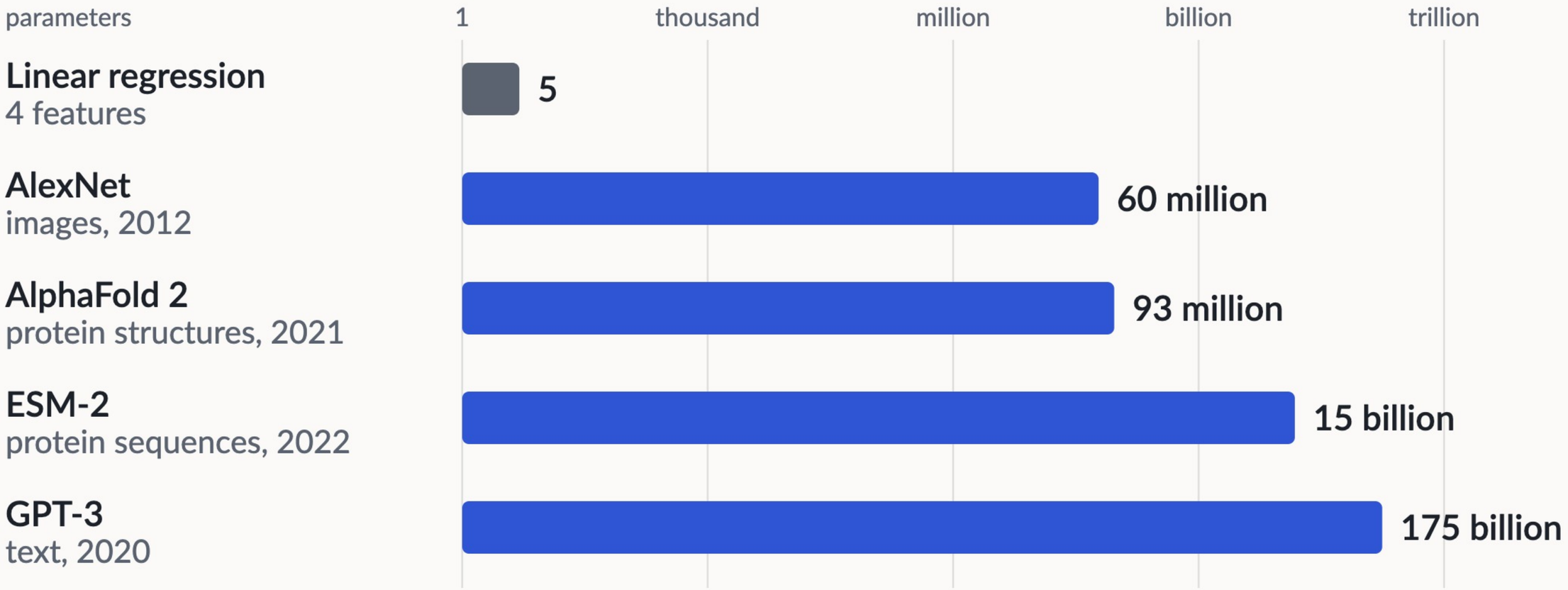
Peptides: just categories



GLP-1, first ten residues

1. **A peptide is a string of letters**
   20 amino acids: a fixed alphabet
2. **One-hot encoding**
3. **Fixed length: a table again**
   10 positions × 20 = 200 numbers

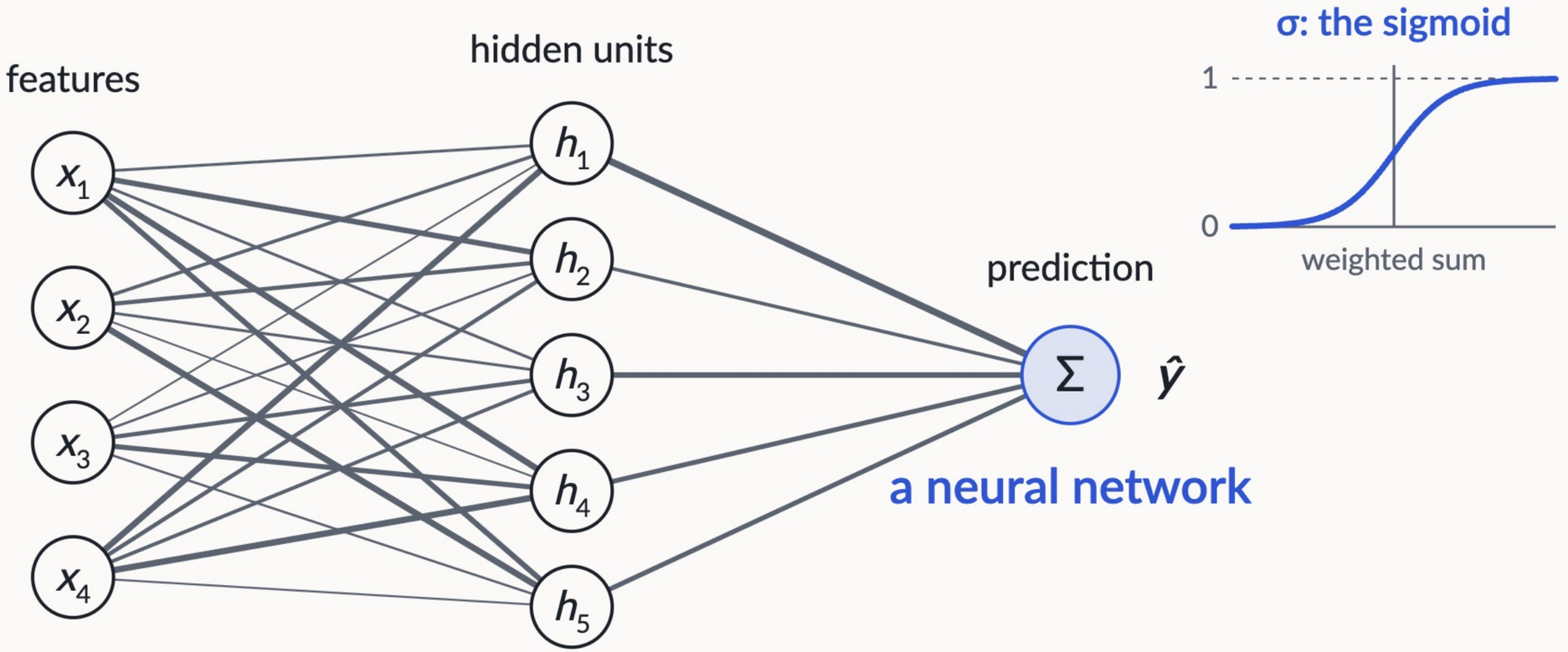
Models have become very large



More parameters: room for more complex relationships. The price is data.

From linear regression to a neural network

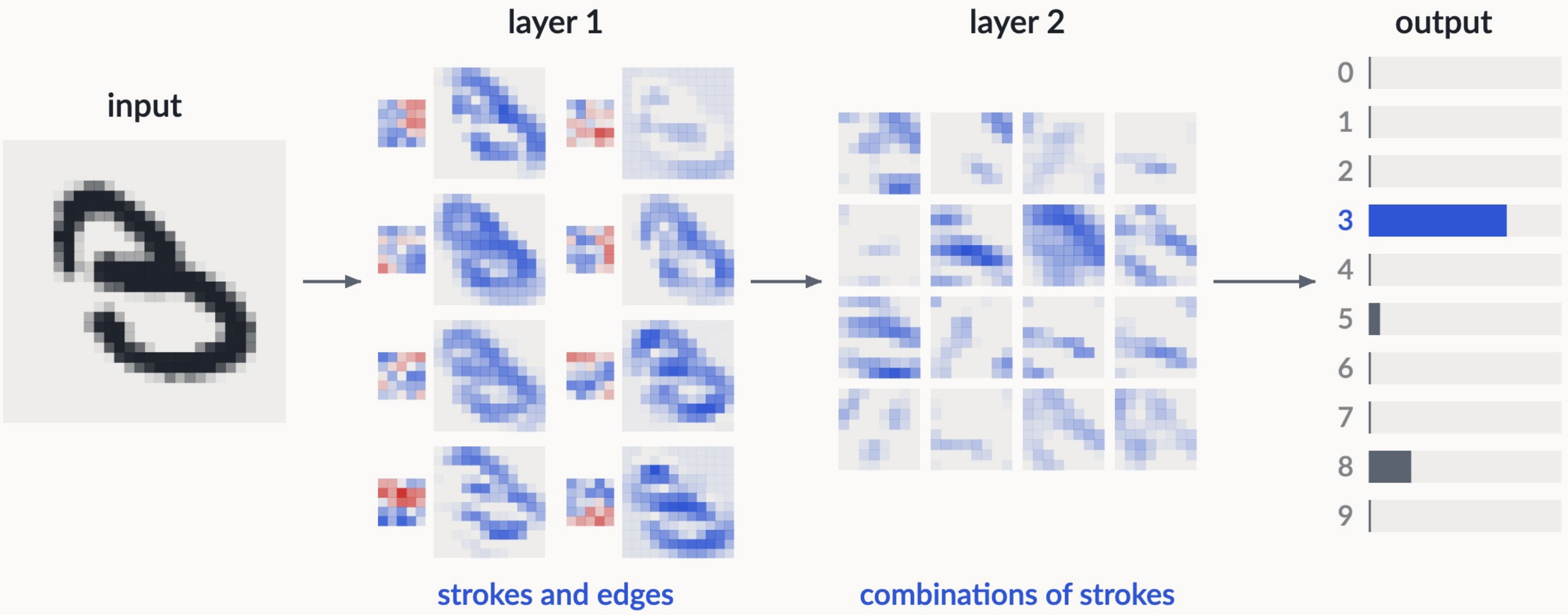
$$h_j = \sigma(w_{j1}x_1 + \ldots + w_{j4}x_4 + b_j) \quad \text{then} \quad \hat{y} = v_1h_1 + \ldots + v_5h_5 + c$$



Each hidden unit: a weighted sum, then a non-linear function σ.

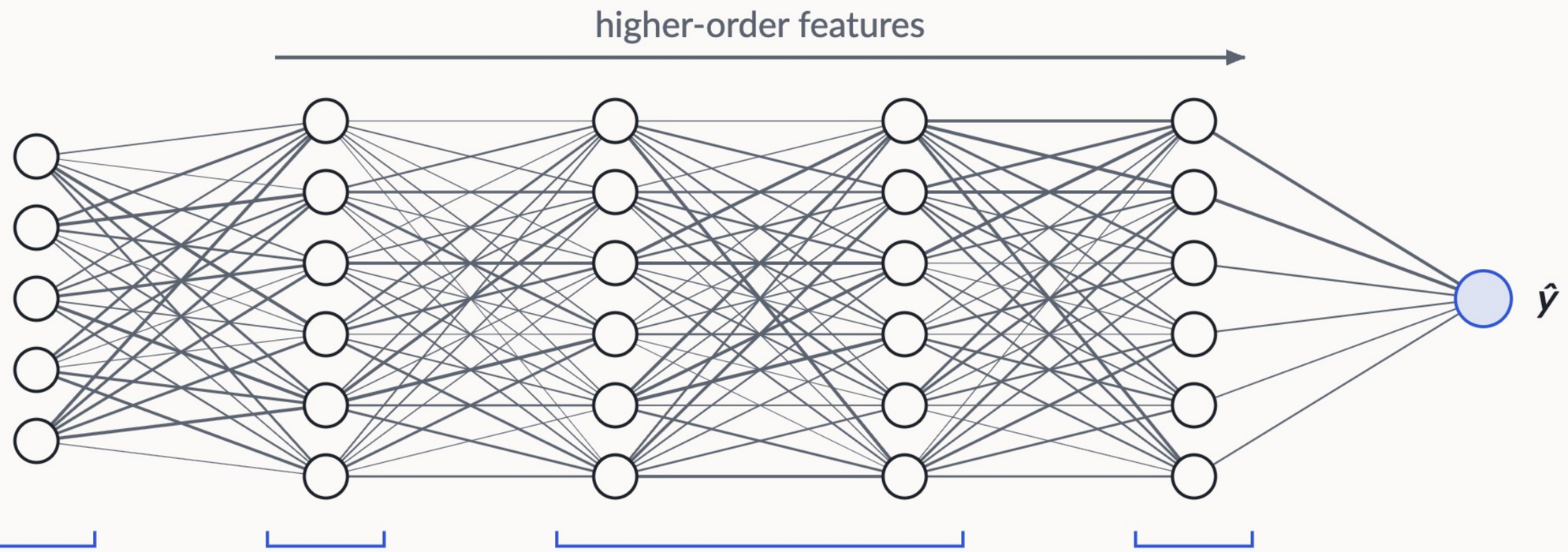
On images: a network learns its own features

trained · test accuracy 97%



Another digit: other features respond.

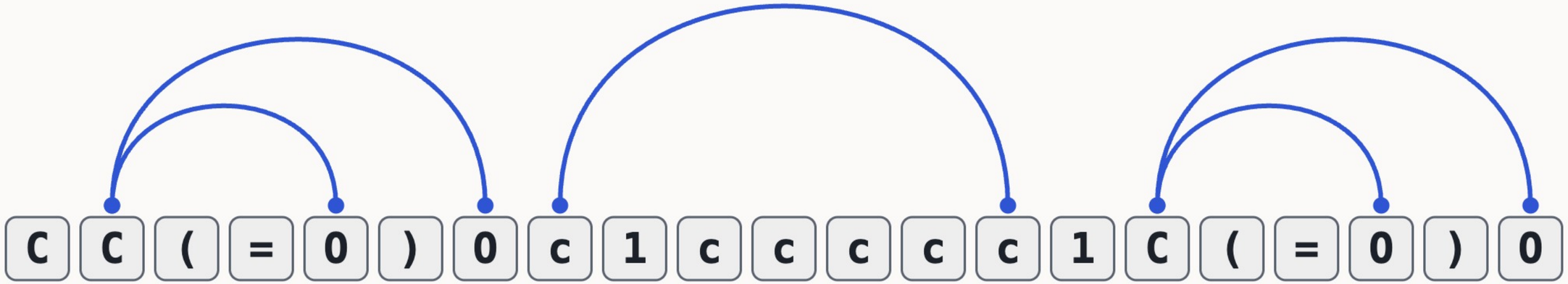
The same for molecules: features on features



Nobody designs these features. They are learned from data.

What a large model has to learn

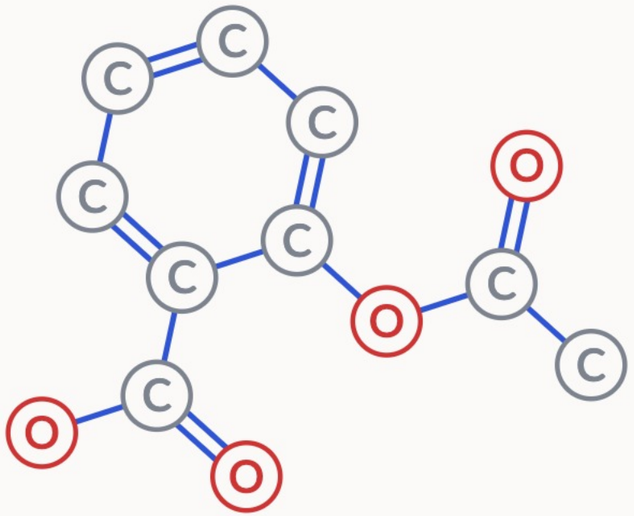
bonded in the molecule, far apart in the string



aspirin, as the model receives it

Nothing in the input says so: the model learns these dependencies from data.

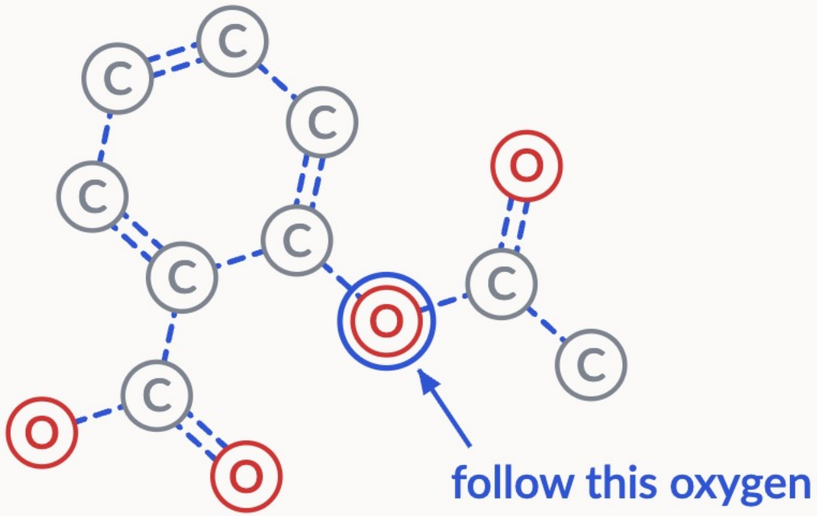
A molecule is a graph



atom	element	hydrogens	aromatic
methyl carbon	C	3	no
ester oxygen	O	0	no
ring carbon	C	0	yes

1. **A SMILES spells out a graph**
2. **Atoms become nodes**
   element, charge, hydrogens, aromatic
3. **Bonds become edges**
   single, double, aromatic
4. **Nothing is thrown away**

Message passing: one round



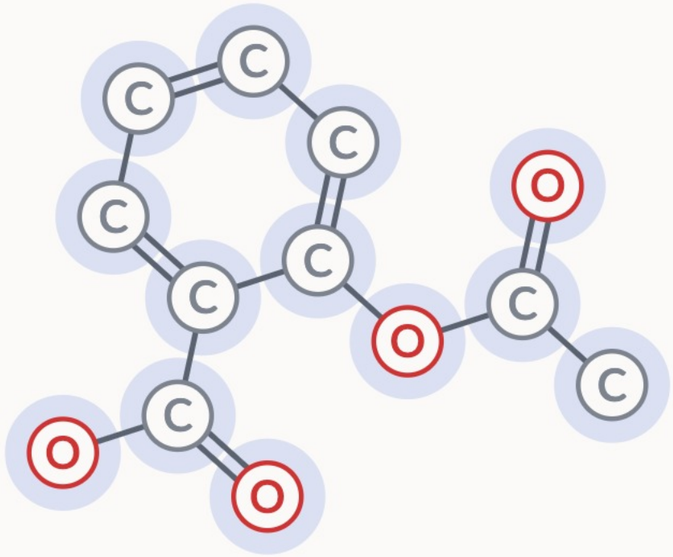
		C	O	aromatic	H
	this oxygen	0	1	0	0
	its neighbour	1	0	0	0
	its neighbour	1	0	1	0
Σ	sum	2	0	1	0

$$\text{new} = \sigma(\ W_1 \cdot \text{own} + W_2 \cdot \text{sum}\ )$$

new vector

Every atom does this at once, with the same weights: one round.

Round after round: a graph neural network



1. **1 round: it knows its neighbours**
2. **2 rounds: two bonds out**
3. **3 rounds: three bonds out**

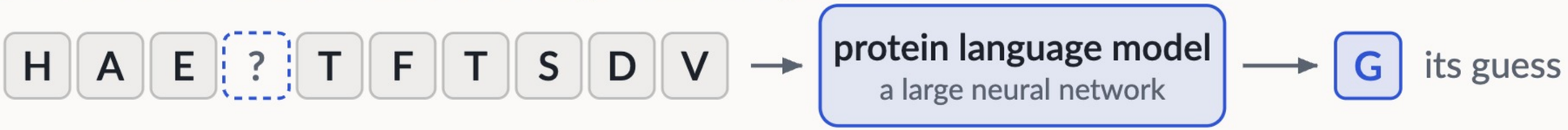
   like a fingerprint, but learned
4. **Add up all atoms, then predict**

   any molecule: one vector of fixed size

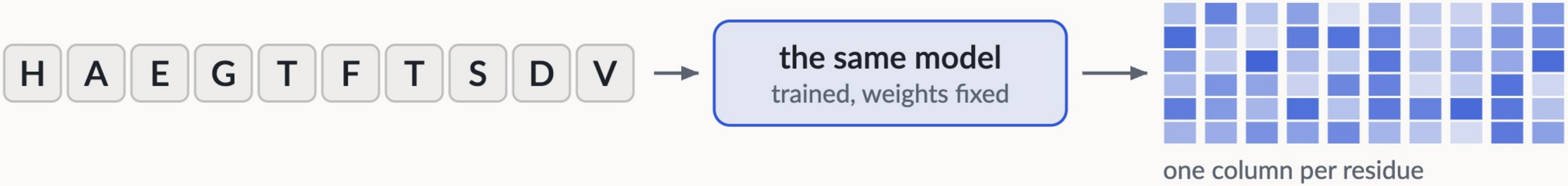


Embeddings from pretrained models

1 Pretrain hide a letter, let a large network guess it



2 Embed a vector of learned features per residue

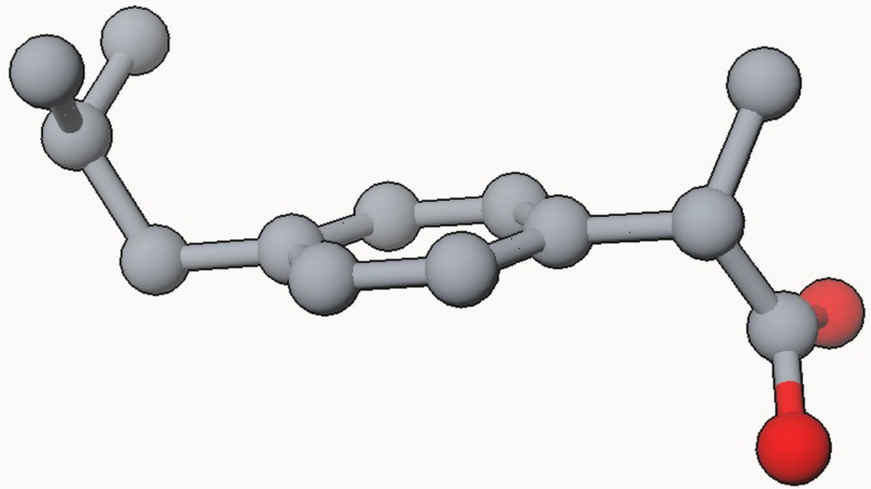


3 Use average: one row per peptide



Features learned from unlabelled sequences: reusable when your data is small.

What about 3D structure?



Drag to turn.

1 **Molecules are 3D objects**

binding is about shape

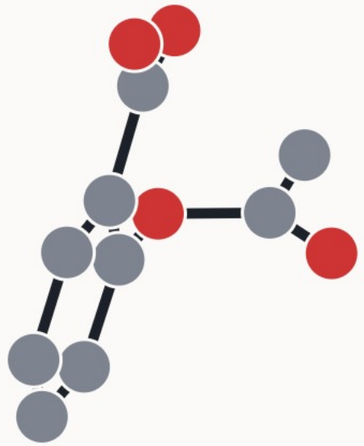
2 **One graph, many shapes**

two conformers of ibuprofen

3 **The graph cannot tell them apart**

nor can SMILES or fingerprints

Simple version: add the 3D coordinates



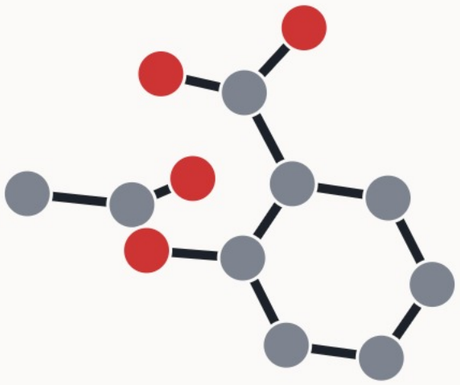
atom	*x*	*y*	*z*
C1	1.9	0.8	−2.9
C2	1.5	0.1	−1.7
O3	2.2	−0.4	−0.8
O4	0.1	0.1	−1.6
C5	−0.4	−0.5	−0.4
C6	−0.8	−1.8	−0.5
C7	−1.4	−2.5	0.6
…		…	

1. **Each atom is a node**
2. **Add x, y, z as features**

   now the network sees the shape
3. **Turn it: every number changes**

   same molecule, different input

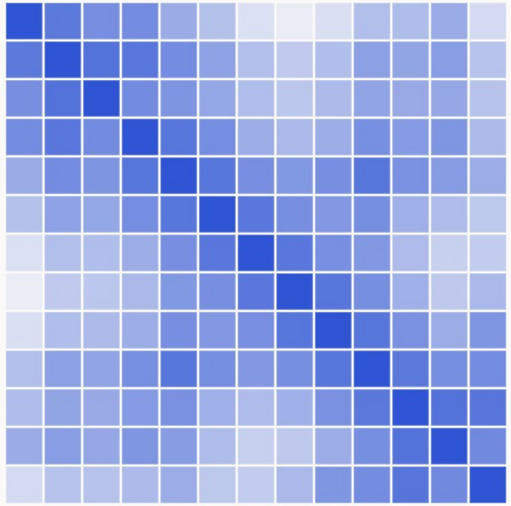
Two ways out



atom	*x*	*y*	*z*
C1	−3.1	0.1	−1.7
C2	−1.6	0.0	−1.5
O3	−0.8	0.4	−2.3
O4	−1.4	−0.7	−0.3
C5	−0.1	−0.8	0.1
C6	0.5	−2.0	−0.2
C7	1.9	−2.2	0.1
…		…	

1 Make the representation invariant

distances between atoms, not positions



all pairs
of atoms:

it does not move

2 Make the network invariant

built so that turning the input cannot change the answer

Two take-aways

1 **Preprocessing turns a raw molecule into a format the model can read**

descriptors · fingerprints · SMILES tokens · graphs · 3D coordinates

2 **More data, fewer handcrafted features**

let the model learn the higher-order features from data itself

> *"… general methods that leverage computation are ultimately the most effective, and by a large margin."*
>
> Rich Sutton, The Bitter Lesson (2019)

Questions?

Next: Lecture 14, predictive ML